

PowerWise Electricity Analyzer

Database Design Guide

Database for the **PowerWise Electricity Analyzer**

It will help to manage the following functionalities:

- User details
- Appliance details
- Electricity consumption details
- Daily and monthly usage records
- Electricity bill calculation
- Energy consumption analysis
- High consumption alerts
- Energy-saving suggestions
- Consumption history
- Usage reports and graphs

We will use **MySQL** as the DBMS to create the database and perform its related operations.

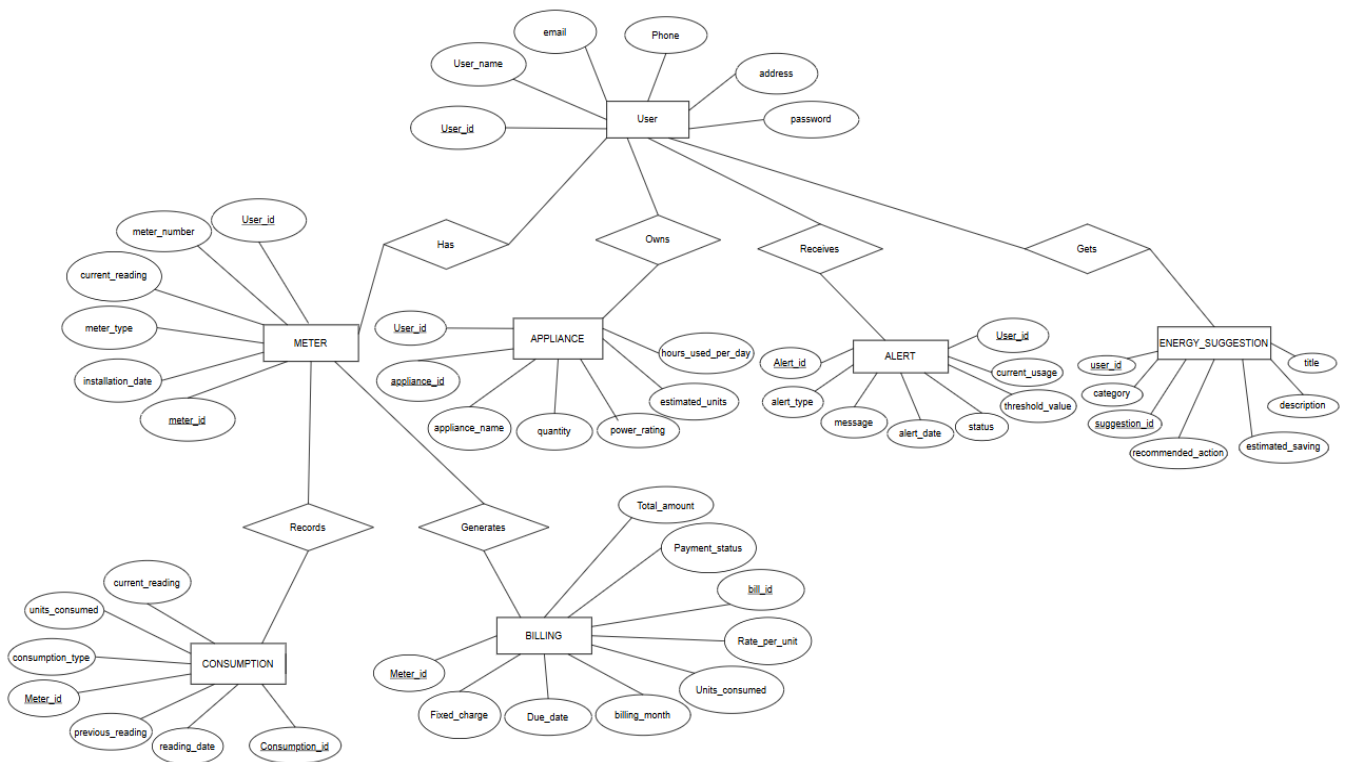
1. E-R Diagram (ERD)

An Entity-Relationship Diagram represents the entities, attributes, relationships, and cardinality of the TravelSafe database.

Identify Entities

The main entities of the PowerWise Electricity Analyzer application are:

1. User
 2. Meter
 3. Appliance
 4. Alert
 5. Energy_suggestion
 6. Consumption
 7. Billing
-



Define Attributes and Relationships

1. User

Attributes:

- User_id (Primary Key)
- User_name
- Email
- Password
- Phone
- Address

Relationships:

- One User can have many Meters. **(One-to-Many)**
- One User can have many Appliances. **(One-to-Many)**
- One User can have many Alerts. **(One-to-Many)**
- One User can have many Energy Suggestions. **(One-to-Many)**

2. Meter

Attributes:

- Meter_id (Primary Key)
- User_id (Foreign Key)
- Meter_number
- Meter_type
- Installation_date
- Current_reading

Relationships:

- One Meter belongs to one User. **(Many-to-One)**
 - One Meter can have many Consumption records. **(One-to-Many)**
-

3. Appliance

Attributes:

- Appliance_id (Primary Key)
- User_id (Foreign Key)
- Appliance_name
- Quantity
- Power_rating
- Estimated_units
- Hours_used_per_day

Relationships:

- One Appliance belongs to one User. **(Many-to-One)**
-

4. Consumption

Attributes:

- ConsumptionID (Primary Key)
- MeterID (Foreign Key)
- PreviousReading
- CurrentReading
- UnitsConsumed
- ConsumptionType
- ReadingDate

Relationships:

- Many Consumption records belong to one Meter. **(Many-to-One)**
-

5. Billing

Attributes:

- Bill_id (Primary Key)
- Meter_id (Foreign Key)
- Units_consumed
- Due_date
- Fixed_charge
- Billing_month
- Rate_per_unit
- Payment_status
- Total_amount

Relationships:

- Many Billing records belong to one Appliance. **(Many-to-One)**
-

6. Alert

Attributes:

- Alert_id (Primary Key)
- User_id (Foreign Key)
- Alert_type
- Alert_date
- Message
- Threshold_value
- Current_usage
- Status

Relationships:

- Many Alerts belong to one User. (Many-to-One)
-

7. Energy Suggestion

Attributes:

- Suggestion_id (Primary Key)
- User_id (Foreign Key)
- Category
- Title
- Description
- Recommended_action
- Estimated_saving

Relationships:

- Many Energy Suggestions belong to one User. (**Many-to-One**)
-

2. Creating a Database

Using MySQL Server, create a new database for the PowerWise Electricity Analyzer application.

```
CREATE DATABASE powerwise;
```

3. Using a Database

Before performing operations, select the database:

```
USE powerwise;
```

4. Creating Tables for Each Entity

```
mysql> show tables;
+-----+
| Tables_in_powerwise |
+-----+
| alert                |
| appliance            |
| billing              |
| consumption          |
| energy_suggestion    |
| meter                |
| users                |
+-----+
7 rows in set (0.02 sec)
```

1. User Table

```
CREATE TABLE users (  
  user_id INT AUTO_INCREMENT PRIMARY KEY,  
  user_name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  phone VARCHAR(15),  
  address VARCHAR(255),  
  password VARCHAR(255) NOT NULL  
);
```

```
mysql> desc users;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| user_id | int | NO | PRI | NULL | auto_increment |  
| user_name | varchar(100) | NO | | NULL | |  
| email | varchar(100) | NO | UNI | NULL | |  
| phone | varchar(15) | YES | | NULL | |  
| address | varchar(255) | YES | | NULL | |  
| password | varchar(255) | NO | | NULL | |  
+-----+-----+-----+-----+-----+-----+  
6 rows in set (0.00 sec)
```

2. Meter Table

```
CREATE TABLE meter (  
  meter_id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT NOT NULL,  
  meter_number VARCHAR(50) UNIQUE NOT NULL,  
  meter_type VARCHAR(50),  
  installation_date DATE,  
  current_reading DECIMAL(10,2) DEFAULT 0.00,  
  FOREIGN KEY (user_id) REFERENCES users(user_id)  
);
```

```
mysql> desc meter;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| meter_id | int | NO | PRI | NULL | auto_increment |  
| user_id | int | NO | MUL | NULL | |  
| meter_number | varchar(50) | NO | UNI | NULL | |  
| meter_type | varchar(50) | YES | | NULL | |  
| installation_date | date | YES | | NULL | |  
| current_reading | decimal(10,2) | YES | | 0.00 | |  
+-----+-----+-----+-----+-----+-----+  
6 rows in set (0.00 sec)
```

3. Appliance Table

```
CREATE TABLE appliance (  
  appliance_id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT NOT NULL,  
  appliance_name VARCHAR(100) NOT NULL,  
  quantity INT DEFAULT 1,  
  power_rating DECIMAL(10,2),  
  hours_used_per_day DECIMAL(5,2),  
  estimated_units DECIMAL(10,2),  
  FOREIGN KEY (user_id) REFERENCES users(user_id)  
);
```

```
mysql> desc appliance;  
+-----+-----+-----+-----+-----+-----+  
| Field          | Type          | Null | Key | Default | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| appliance_id   | int           | NO   | PRI | NULL    | auto_increment |  
| user_id        | int           | NO   | MUL | NULL    |                |  
| appliance_name | varchar(100)  | NO   |     | NULL    |                |  
| quantity       | int           | YES  |     | 1       |                |  
| power_rating   | decimal(10,2) | YES  |     | NULL    |                |  
| hours_used_per_day | decimal(5,2) | YES  |     | NULL    |                |  
| estimated_units | decimal(10,2) | YES  |     | NULL    |                |  
+-----+-----+-----+-----+-----+-----+  
7 rows in set (0.00 sec)
```

4. Consumption Table

```
CREATE TABLE consumption (  
  consumption_id INT AUTO_INCREMENT PRIMARY KEY,  
  meter_id INT NOT NULL,  
  previous_reading DECIMAL(10,2),  
  current_reading DECIMAL(10,2),  
  units_consumed DECIMAL(10,2),  
  consumption_type VARCHAR(50),  
  reading_date DATE,  
  FOREIGN KEY (meter_id) REFERENCES meter(meter_id)  
);
```

```
mysql> desc consumption;  
+-----+-----+-----+-----+-----+-----+  
| Field          | Type          | Null | Key | Default | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| consumption_id | int           | NO   | PRI | NULL    | auto_increment |  
| meter_id       | int           | NO   | MUL | NULL    |                |  
| previous_reading | decimal(10,2) | YES  |     | NULL    |                |  
| current_reading | decimal(10,2) | YES  |     | NULL    |                |  
| units_consumed  | decimal(10,2) | YES  |     | NULL    |                |  
| consumption_type | varchar(50)   | YES  |     | NULL    |                |  
| reading_date    | date          | YES  |     | NULL    |                |  
+-----+-----+-----+-----+-----+-----+  
7 rows in set (0.00 sec)
```

5. Billing Table

```
CREATE TABLE billing (  
  bill_id INT AUTO_INCREMENT PRIMARY KEY,  
  meter_id INT NOT NULL,  
  units_consumed DECIMAL(10,2),  
  rate_per_unit DECIMAL(10,2),  
  fixed_charge DECIMAL(10,2),  
  total_amount DECIMAL(10,2),  
  billing_month DATE,  
  due_date DATE,  
  payment_status VARCHAR(30) DEFAULT 'Pending',  
  FOREIGN KEY (meter_id) REFERENCES meter(meter_id)  
);
```

```
mysql> desc billing;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| bill_id    | int           | NO   | PRI | NULL    | auto_increment |  
| meter_id   | int           | NO   | MUL | NULL    |                |  
| units_consumed | decimal(10,2) | YES  |     | NULL    |                |  
| rate_per_unit | decimal(10,2) | YES  |     | NULL    |                |  
| fixed_charge | decimal(10,2) | YES  |     | NULL    |                |  
| total_amount | decimal(10,2) | YES  |     | NULL    |                |  
| billing_month | date          | YES  |     | NULL    |                |  
| due_date    | date          | YES  |     | NULL    |                |  
| payment_status | varchar(30)   | YES  |     | Pending |                |  
+-----+-----+-----+-----+-----+-----+  
9 rows in set (0.00 sec)
```

6. Alert

```
CREATE TABLE alert (  
  alert_id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT NOT NULL,  
  current_usage DECIMAL(10,2),  
  threshold_value DECIMAL(10,2),  
  alert_type VARCHAR(50),  
  message VARCHAR(255),  
  alert_date DATETIME,  
  status VARCHAR(30) DEFAULT 'Unread',  
  FOREIGN KEY (user_id) REFERENCES users(user_id)  
);
```

```
mysql> desc alert;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| alert_id    | int           | NO   | PRI | NULL    | auto_increment |  
| user_id     | int           | NO   | MUL | NULL    |                |  
| current_usage | decimal(10,2) | YES  |     | NULL    |                |  
| threshold_value | decimal(10,2) | YES  |     | NULL    |                |  
| alert_type   | varchar(50)   | YES  |     | NULL    |                |  
| message      | varchar(255)  | YES  |     | NULL    |                |  
| alert_date   | datetime      | YES  |     | NULL    |                |  
| status       | varchar(30)   | YES  |     | Unread  |                |  
+-----+-----+-----+-----+-----+-----+  
8 rows in set (0.00 sec)
```

7. Energy Suggestion

```
CREATE TABLE energy_suggestion (  
  suggestion_id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT NOT NULL,  
  title VARCHAR(150),  
  category VARCHAR(100),  
  description TEXT,  
  recommended_action TEXT,  
  estimated_saving DECIMAL(10,2),  
  FOREIGN KEY (user_id) REFERENCES users(user_id)  
);
```

```
mysql> desc energy_suggestion;
```

Field	Type	Null	Key	Default	Extra
suggestion_id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	MUL	NULL	
title	varchar(150)	YES		NULL	
category	varchar(100)	YES		NULL	
description	text	YES		NULL	
recommended_action	text	YES		NULL	
estimated_saving	decimal(10,2)	YES		NULL	

```
7 rows in set (0.00 sec)
```

